Your Submission Report

Welcome to your submission report. This document contains the details of your submission.

Report metadata:

Generated on: 2025-08-06 14:18:15

Ground truth file: data/cGround_Truth_data_10trials_150stims_180000ms.hdf5

Submission file: data/cNo_Inhibitory_Neurons_10trials_150stims_180000ms.hdf5

Total trials in ground truth: 10

Total trials in submission: 10

Number of neurons in ground truth: 5

Number of neurons in submission: 5

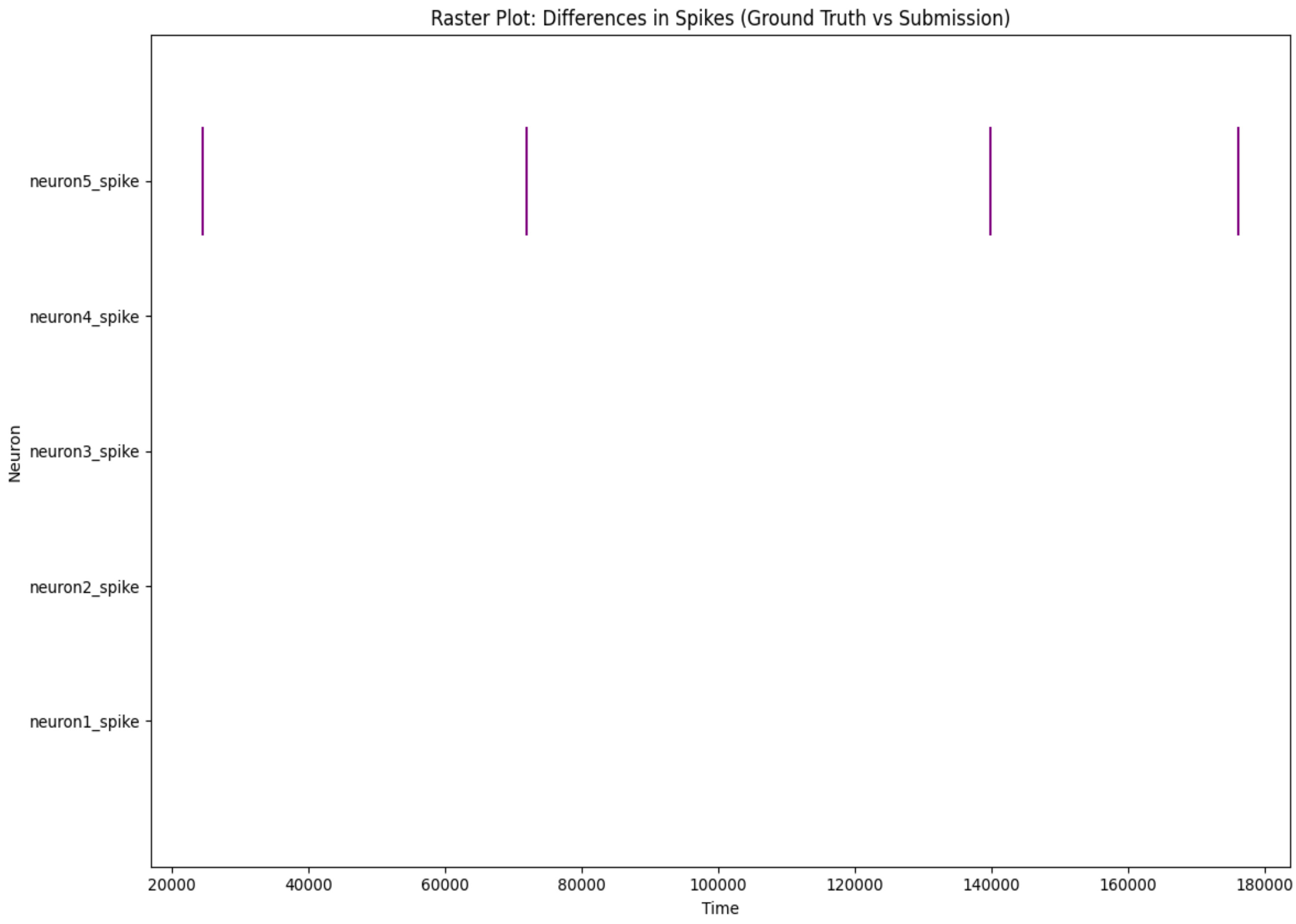
Total number of trials in Ground Truth: 10

Total number of trials in Submission: 10

KS Statistic (Kolmogorov-Smirnov Statistic) The Kolmogorov-Smirnov (KS) statistic measures the maximum distance between the cumulative distribution functions of two datasets. A value of 0.0 typically indicates a perfect match between the distributions, suggesting the model's predictions align perfectly with the actual data. Pearson Correlation The Pearson correlation coefficient measures the linear relationship between two variables, in this case, the predicted and actual values. It ranges from -1 to +1. A value of 1.0 indicates a perfect positive linear relationship, where the predicted and actual values increase together perfectly. RMSE (Root Mean Square Error) The Root Mean Square Error (RMSE) is a measure of the differences between values predicted by a model and the actual observed values. It's the square root of the average of the squared errors. An RMSE of 0.0 signifies that the model's predictions were exactly the same as the actual values, meaning there was no error. Overall Score The Overall Score is a combined metric that evaluates the performance of each Neuron based on the other statistics.

Neuron	KS Statistic	Pearson Correlation	RMSE	Overall Score
Neuron 1	0.0	1.0	0.0	100.0
Neuron 2	0.0	1.0	0.0	100.0
Neuron 3	0.0	1.0	0.0	100.0
Neuron 4	0.0	1.0	0.0	100.0
Neuron 5	2.222e-05	0.9934	5.676e-05	0.0

A raster plot is a fundamental visualization tool in neuroscience for representing spike trains. It displays neural activity across multiple trials or neurons in a compact and intuitive way. In a raster plot, time is typically on the x-axis, and each horizontal line (or row) represents a single trial or a single neuron. A small mark, often a vertical line or a dot, is placed on each row at the precise time a spike occurred. By stacking these rows together, a raster plot allows researchers to easily visualize patterns in spiking activity, such as changes in firing rate over time (e.g., in response to a stimulus), the trial-to-trial variability of a neuron's response, or the synchronized firing of multiple neurons.



A peristimulus time histogram (PSTH) is a common way to visualize neural activity in response to a specific event, or "stimulus." To read a PSTH, you first look at the horizontal axis, which represents **time** relative to the stimulus onset. The stimulus is usually marked at time 0. The vertical axis represents the **firing rate** of a neuron, often in spikes per second, and each bar of the histogram shows how many spikes occurred within a specific time bin. By analyzing the shape of the histogram, you can understand how a neuron's activity changes in response to the stimulus. A sudden increase in the height of the bars after time 0 suggests the neuron is excited by the stimulus, while a decrease suggests it's inhibited.